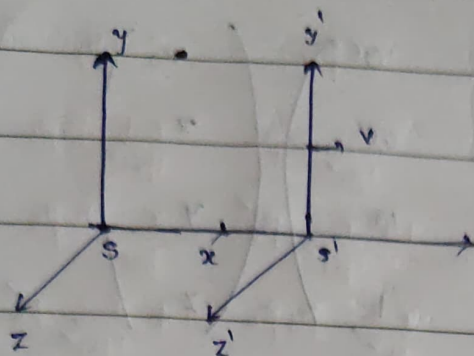


11/09/26

classmate

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$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}$$

$$y' = y$$

$$z' = z$$

$$t' = \frac{t - vx/c^2}{\sqrt{1 - v^2/c^2}}$$

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

$$\beta = \frac{v}{c}$$

$$x' = \gamma(x - vt)$$

$$= \gamma(x - \frac{v}{c}ct)$$

$$= \gamma(x - \beta ct)$$

$$= \gamma(x - \beta \frac{ct}{i}i)$$

$$\vec{x}' = \gamma(\vec{x} + \beta \vec{x}_0 i)$$

$$t' = \gamma(t - \frac{vx}{c^2})$$

$$x'_0 = i c t'$$

$$= i c \gamma(t - \frac{vx}{c^2})$$

$$= \gamma(i c t - \frac{i v x}{c})$$

$$= \gamma(\vec{x}_0 - i \beta \vec{x})$$

$$\begin{pmatrix} x_0' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \gamma & -i\beta\gamma & 0 & 0 \\ i\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_0 \\ x \\ y \\ z \end{pmatrix}$$

$$\boxed{x'_\mu = \Lambda_{\mu\nu} x_\nu} \quad \mu, \nu = 0, 1, 2, 3$$

$$x_1 = x$$

$$x_2 = y$$

$$x_3 = z$$

$$\vec{x} = \vec{x}_\perp + \vec{x}_\parallel$$

$\vec{x}_\perp$ The component of $\vec{x}$ perpendicular to the direction of motion.

$\vec{x}_\parallel$ The component of $\vec{x}$ parallel to the direction of motion.

$$\vec{x}' = \vec{x}_\perp + \frac{\vec{x}_\parallel - \vec{v}t}{\sqrt{1 - v^2/c^2}}$$

$$= (\vec{x} - \vec{x}_\parallel) + (\vec{x}_\parallel - \vec{v}t)\gamma$$

$$= \vec{x} + (\gamma - 1)\vec{x}_\parallel - \gamma\vec{v}t$$

$$\vec{x}_\parallel = |x_\parallel| \hat{e}$$

$$= |x_\parallel| \frac{\vec{v}}{|\vec{v}|} = \frac{\vec{x} \cdot \vec{v}}{|\vec{v}|} \frac{\vec{v}}{|\vec{v}|} = \frac{(\vec{x} \cdot \vec{v})}{|\vec{v}|^2} \vec{v}$$

$$\vec{x}' = \vec{x} + (r-1) \frac{(\vec{x} \cdot \vec{v}) \vec{v}}{|\vec{v}|^2} + r \vec{v} t$$

$$\left(\vec{\beta} = \frac{\vec{v}}{c} \right)$$

$$\vec{x}' = \vec{x} + (r-1) \frac{(\vec{x} \cdot \vec{v})}{c} \frac{\vec{v} c}{|\vec{v}|^2} + r \frac{\vec{v} c t}{c}$$

$$= \vec{x} + (r-1) (\vec{x} \cdot \vec{\beta}) \frac{\vec{\beta} c^2}{|\vec{v}|^2} + r \vec{\beta} \frac{ict}{i}$$

$$\vec{x}' = \vec{x} + (r-1) (\vec{x} \cdot \vec{\beta}) \vec{\beta} \left(\frac{c^2}{|\vec{v}|^2} \right) + i r \vec{\beta} x^0$$

$$x'_1 \hat{i} + x'_2 \hat{j} + x'_3 \hat{k}$$

$$x_1 \hat{i} + x_2 \hat{j} + x_3 \hat{k}$$

$$t' = \frac{t - \vec{v} \cdot \vec{x} / c^2}{\sqrt{1 - \vec{v}^2 / c^2}} = \frac{(t - \vec{\beta} \cdot \vec{x} / c) \gamma}{\sqrt{1 - \vec{v}^2 / c^2}}$$

$$= \gamma \left(\frac{ict - \vec{\beta} \cdot \vec{x}}{c} \right) = \frac{\gamma}{c} (ict - \vec{\beta} \cdot \vec{x})$$

$$= \frac{\gamma}{c} (ix^0 - \vec{\beta} \cdot \vec{x})$$

$$(\vec{\beta} = \beta_1 \hat{i} + \beta_2 \hat{j} + \beta_3 \hat{k})$$

$$ict' = \frac{\gamma}{c} (x^0 - i \vec{\beta} \cdot \vec{x})$$

$$x'_0 = \frac{\gamma}{c} (x^0 - i \vec{\beta} \cdot \vec{x}) = \gamma x_0 - i \gamma \beta_1 x_1 - i \gamma \beta_2 x_2 - i \gamma \beta_3 x_3$$

$$\begin{pmatrix} x'_0 \\ x'_1 \\ x'_2 \\ x'_3 \end{pmatrix} = \begin{pmatrix} \gamma & -i\gamma\beta_1 & -i\gamma\beta_2 & -i\gamma\beta_3 \\ i\gamma\beta_1 & 1 + \frac{\gamma-1}{|\beta|^2} \beta_1^2 & \frac{\gamma-1}{|\beta|^2} \beta_1\beta_2 & \frac{\gamma-1}{|\beta|^2} \beta_1\beta_3 \\ i\gamma\beta_2 & \frac{\gamma-1}{|\beta|^2} \beta_1\beta_2 & 1 + \frac{\gamma-1}{|\beta|^2} \beta_2^2 & \frac{\gamma-1}{|\beta|^2} \beta_2\beta_3 \\ i\gamma\beta_3 & \frac{\gamma-1}{|\beta|^2} \beta_1\beta_3 & \frac{\gamma-1}{|\beta|^2} \beta_2\beta_3 & 1 + \frac{\gamma-1}{|\beta|^2} \beta_3^2 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

when the velocity is along x-axis only then it turns into ordinary Lorentz

Transform ($\beta_1 = \beta$, $\beta_2 = \beta_3 = 0$)

$$X_\mu \rightarrow X'_\mu = \Lambda_{\mu\nu} X^\nu$$

$$X^\mu \rightarrow X'^\mu = \Lambda^{\mu\nu} X_\nu$$

$$= \Lambda^\mu_\nu X^\nu$$

$$\tilde{X}^2 = X_\mu X^\mu$$

↓
Lorentz scalar

$$\begin{aligned} &= X^0 X_0 + X^1 X_1 + X^2 X_2 + X^3 X_3 \\ &= X^0 X^0 - X^1 X^1 - X^2 X^2 - X^3 X^3 \\ &= X^{0^2} - X^{1^2} - X^{2^2} - X^{3^2} \\ &= -X^{0^2} + X^{1^2} + X^{2^2} + X^{3^2} \end{aligned}$$

* imp

Inert a metric

$$\eta_{\mu\nu} = \begin{pmatrix} +1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

↓
Minkowski matrix

$$g = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & +1 & 0 & 0 \\ 0 & 0 & +1 & 0 \\ 0 & 0 & 0 & +1 \end{pmatrix}$$

metric

4x4 matrix.